

DTS224: Database Management System 1 Test1 (Type A) Time: 45mins

- (a) Explain the following entity relationship concepts with example :
(i) Attribute (ii) Ternary relationship (iii) Entity
- (b) Express the relationships in the business rule with just one connected ER diagram to show the degree and cardinalities of the relationships.
- A hospital has a large number of registered physicians. Attributes of PHYSICIAN include Physician ID (the identifier) and Specialty. Patients are admitted to the hospital by physicians. Attributes of PATIENT include Patient ID (the identifier) and Patient Name. Any patient who is admitted must have exactly one admitting physician. A physician may optionally admit any number of patients. Once admitted, a given patient must be treated by at least one physician. A particular physician may treat any number of patients, or may not treat any patients. Whenever a physician treats a patient, the hospital wishes to record the treatment details. Components of Treatment Detail include Date, Time, and Results. Did you draw more than one relationship between physician and patient? How would you include in the ERD the need to represent the date a patient is admitted for each time he or she is admitted? A patient is assigned to a particular ward, which can accommodate many patients at a time or is vacant. A ward is characterized by its name and type.

DTS224: Database Management System 1 Test1 (Type B) Time: 45mins

- (a) Explain the following entity relationship concepts with example:
(i) Multivalued attribute (iii) Associative Entity (iv) Binary relationship
- (b) Provide good definitions for all relationships and use an ERD to model this situation:
- A laboratory has several chemists who work on one or more projects. Chemists also may use certain kinds of equipment on each project. Attributes of CHEMIST include Employee ID (identifier), Name, and Phone No. Attributes of PROJECT include Project ID (identifier) and Start Date. Attributes of EQUIPMENT include Serial No and Cost. The organization wishes to record Assign Date—that is, the date when a given equipment item was assigned to a particular chemist working on a specified project. A chemist must be assigned to at least one project and one equipment item. A given equipment item need not be assigned, and a given project need not be assigned either a chemist or an equipment item.

DTS224: Database Management System 1 Test1 (Type C) Time: 45mins

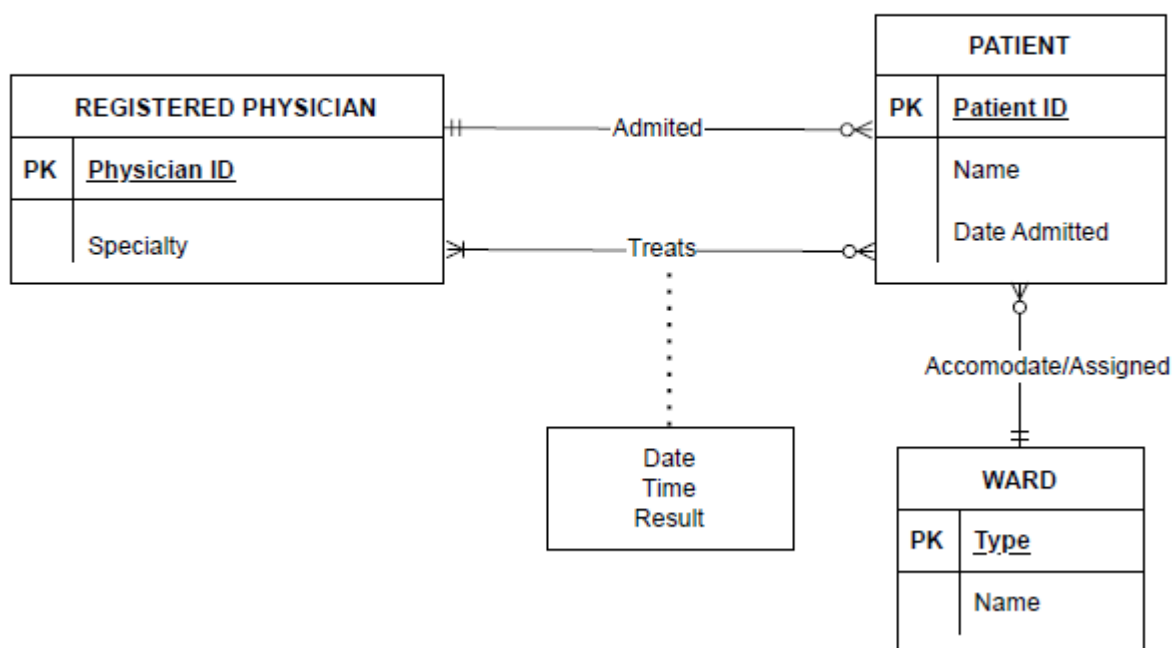
- (a) Explain the following entity relationship concepts with example:
(i) Weak Entity Type (iii) Strong Entity Type (iv) Unary relationship
- (b) Prepare an ERD for a real estate firm that lists property for sale.
- The firm has a number of sales offices in several states. Attributes of sales offices include Office_Number (identifier) and Location. Each sales office is assigned one or more employees. Attributes of employees include Employee_ID (identifier) and Employee_Name. An employee must be assigned to only one sales office. For each sales office, there is always one employee assigned to manage that office. An employee may manage only the sales office to which they are assigned. The firm lists property for sale. Attributes of property include Property_ID (identifier) and Location. Components of Location include Address, City, State, and Zipcode. Each unit of property must be listed with one (and only one) of the sales offices. A sales office may have any number of properties lists or may have no properties listed. Each unit of property has one or more owners. Attributes of owners include Owner_ID (identifier) and Owner_Name. An owner may own one or more units of property. An attribute of the relationship between property and owner is Percent_Owned.

SOLUTION

TYPE A

- (i) An attribute is a **property or characteristic of an entity type** that is of interest to the organization. (2 marks)
- (ii) A ternary relationship is a *simultaneous* relationship among the instances of three entity types. (2 marks)
- (iii) An entity is a **person, a place, an object, an event, or a concept in the user environment about which the organization wishes to maintain data.** (2 marks)

(b) Modelling Solution



- a. Yes, two relationships were drawn between Registered Physicians and patients. One of the relationships represents Admitted, and the other one represents Treats.
- b. The date a patient is admitted is captured as an attribute for the patient.

Note: 2 marks per entity type = 6 marks

1 mark per relationship or Attribute on relationship = 4 marks

Total Mark = (16/16) * 10

TYPE B

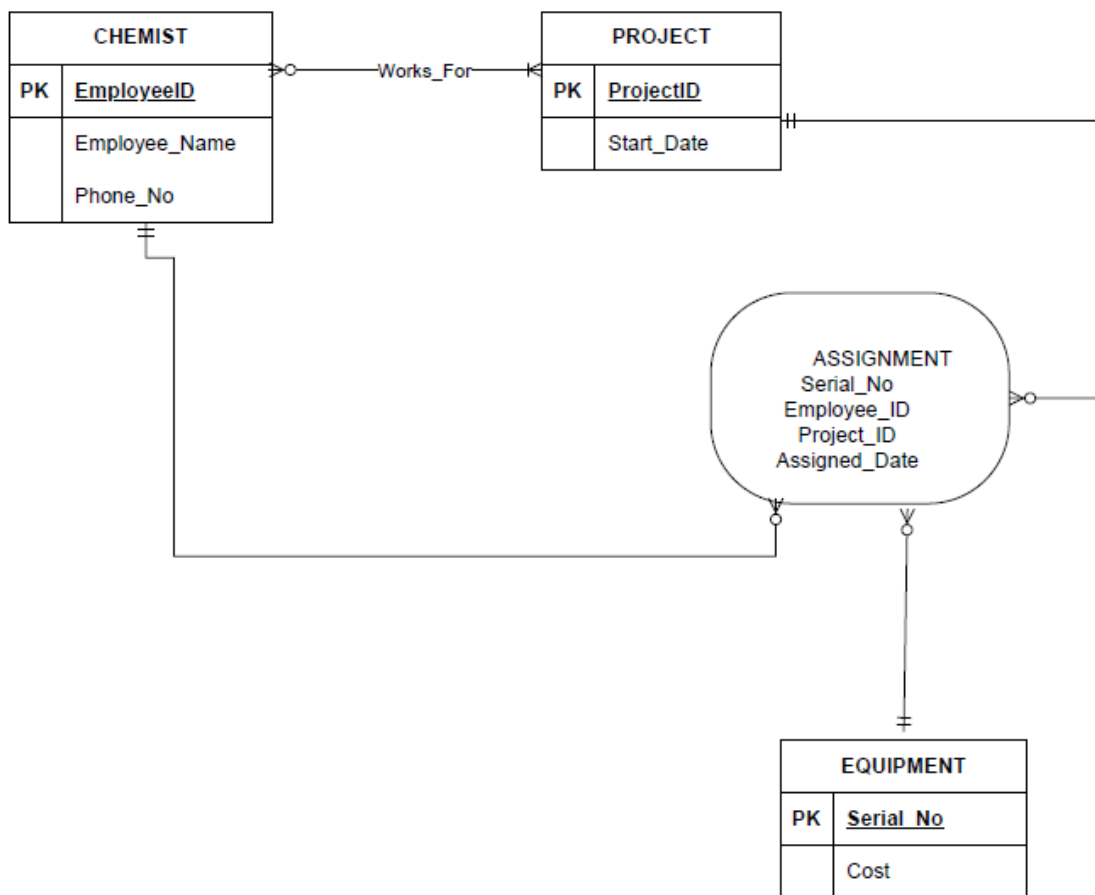
- (i) A multivalued attribute is an **attribute that may take on more than one value for a given entity (or relationship) instance.** Multivalued attribute with curly brackets around the attribute.

(2 marks)

- (ii) An associative entity is **an entity type that associates the instances of one or more entity types and contains attributes peculiar to the relationship between those entity instances. (2 marks)**
- (iii) A binary **relationship** is a relationship between the instances of two entity types and is the most common type of relationship encountered in data modeling. **(2 marks)**

(b)Modelling

Solution



Note: 2 marks per entity type = 8 marks

1 mark per relationship or Attribute on relationship = 4 marks

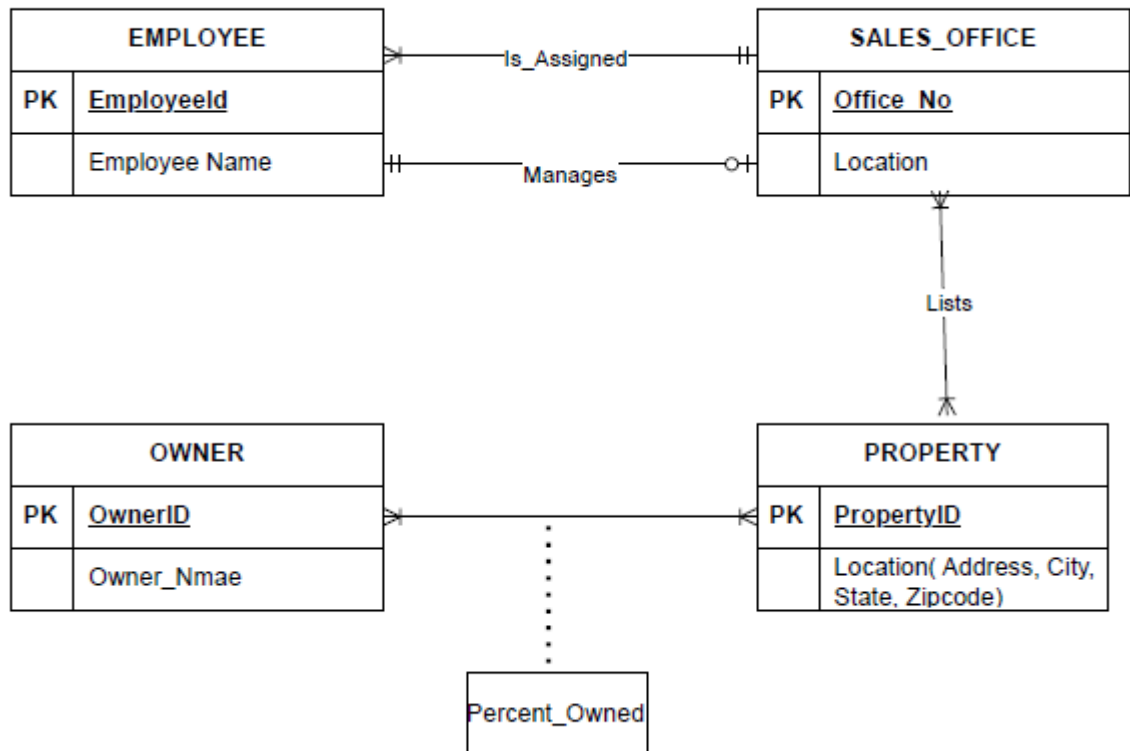
Total Mark = (18/18) * 10

TYPE C

- (i) A weak entity type is **an entity type whose existence depends on some other entity type**. It is also referred to as a *dependent entity*. A weak entity type has no business meaning in an E-R diagram without the entity on which it depends **(2 marks)**

- (ii) Strong entity type is **one that exists independently of other entity types**. It can also be referred to as independent entity. (2 marks)
- (iii) A unary relationship is **a relationship between the instances of a *single* entity type**. Unary relationships are also called *recursive relationships*. (2 marks)

(b) Modelling Solution



Note: 2 marks per entity type = 8 marks

1 mark per relationship or Attribute on relationship = 5 marks

Total Mark = (19/19) * 10